

Dividing fractions



1 Simplify:

a $\frac{1}{8} \div \frac{1}{5}$

e $\frac{1}{10} \div \frac{1}{3}$

b $\frac{1}{2} \div \frac{9}{10}$

f $\frac{1}{2} \div \frac{1}{9}$

c $\frac{1}{5} \div \frac{3}{7}$

g $\frac{2}{3} \div \frac{1}{8}$

d $\frac{3}{4} \div \frac{1}{9}$

h $\frac{1}{11} \div \frac{3}{7}$

2 Simplify:

a $\frac{2}{7} \div \frac{5}{3}$

e $\frac{28}{9} \div \frac{4}{9}$

i $\frac{2}{3} \div \frac{11}{15}$

m $\frac{2}{36} \div \frac{4}{7}$

b $\frac{9}{10} \div \frac{10}{9}$

f $\frac{3}{15} \div \frac{15}{4}$

j $\frac{10}{7} \div \frac{10}{7}$

n $\frac{3}{55} \div \frac{7}{20}$

c $\frac{2}{7} \div \frac{7}{18}$

g $\frac{21}{27} \div \frac{3}{2}$

k $\frac{4}{5} \div \frac{36}{35}$

o $\frac{10}{99} \div \frac{45}{22}$

d $\frac{3}{15} \div \frac{4}{3}$

h $\frac{5}{7} \div \frac{45}{2}$

i $\frac{3}{7} \div \frac{14}{27}$

p $\frac{22}{27} \div \frac{3}{2}$

3 Simplify:

a $3 \div \frac{1}{11}$

e $\frac{3}{5} \div 4$

b $\frac{1}{9} \div 8$

f $\frac{65}{9} \div 5$

c $\frac{4}{3} \div 5$

g $3 \div \frac{39}{10}$

d $\frac{6}{11} \div 8$

h $7 \div \frac{98}{12}$

4 Simplify:

a $1\frac{2}{5} \div \frac{3}{4}$

e $2\frac{1}{3} \div 6\frac{2}{12}$

b $2\frac{1}{4} \div 1\frac{2}{5}$

f $3\frac{3}{8} \div 2\frac{1}{4}$

c $1\frac{2}{7} \div 3\frac{1}{2}$

g $6\frac{3}{4} \div 4\frac{2}{7}$

d $3\frac{1}{3} \div 2\frac{1}{2}$

h $8\frac{1}{2} \div 1\frac{5}{6}$

5 Simplify:

a $\frac{1}{5} \times \frac{1}{3} \div \frac{1}{2}$

e $\frac{1}{7} \times \frac{2}{9} \div \frac{4}{27}$

b $\frac{2}{3} \div \frac{1}{9} \times \frac{2}{5}$

f $\frac{1}{7} \times \frac{5}{8} \div \frac{15}{49}$

c $\frac{3}{6} \div \frac{12}{5} \div \frac{1}{2}$

g $\frac{18}{25} \div \frac{9}{10} \times \frac{1}{3}$

d $\frac{3}{4} \div \frac{2}{5} \times \frac{1}{6}$

h $\frac{8}{11} \div \frac{2}{3} \times \frac{15}{2}$

Applications

6 A family drinks $\frac{3}{4}$ of a carton of milk per day. How many days will 6 cartons of milk last the family?

- 7 A glass holds $\frac{3}{8}$ of a litre of lemonade. How many glasses of lemonade can be filled with 6 litres of lemonade?
- 8 Yang eats $\frac{2}{5}$ of a packet of biscuits each day. If he buys 12 packets, how many days will the packets last?
- 9 If $\frac{2}{3}$ of a tart is shared equally between 5 people, what fraction of the tart would each person receive?
- 10 A truck tank contains 42 L when $\frac{3}{4}$ full. What is the capacity of the tank when full?
- 11 If a garden snail can move $\frac{1}{100}$ km/h, how many hours would it take for the garden snail to reach $\frac{3}{4}$ km?
- 12 Christine has $1\frac{3}{4}$ of a metre of ribbon. She uses $\frac{7}{20}$ to wrap a present. How many presents can she wrap with her ribbon?
- 13 Sheldon is planning to sell banana bread on the upcoming school fair. If a loaf requires $\frac{3}{4}$ cups of sugar, how many banana bread loaves can Sheldon make if he has $8\frac{2}{8}$ cups of sugar on hand?
- 14 Jean is making bows for her gift wrapping business. A bow requires $6\frac{1}{6}$ inches of ribbon. If she has $55\frac{1}{2}$ inches of ribbon, how many bows can Jean make?
- 15 Benjamin wants to engrave his name on a metal plate. There is $1\frac{2}{4}$ cm space on the plate. There are three sizes of letters to choose from: $\frac{1}{8}$ cm wide, $\frac{1}{6}$ cm wide and $\frac{1}{4}$ cm wide.
- a Find the maximum number of letters that Benjamin can engrave on the plate using the following sizes:
- i $\frac{1}{8}$ cm wide ii $\frac{1}{6}$ cm wide iii $\frac{1}{4}$ cm wide
- b Which letter sizes could he use for his name?